

SLIDE 1: TITLE & PRESENTATION DETAILS

B.Sc. Graduation Project Defense Presentation

- **Project Title:** DESIGN AND IMPLEMENTATION OF AN INTELLIGENT DOCUMENT ANALYSIS AND LEARNING ASSISTANT SYSTEM (AuraQA)
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- **Degree & Session:** Bachelor of Science (B.Sc.) Computer Science Defense Presentation (July, 2026)

SLIDE 2: SHORT ABSTRACT

Executive Summary of the Research Project

- **Problem Context:** Cloud-based LLMs present severe data privacy risks and high operational API costs when processing private academic documents.
- **Proposed Solution:** AuraQA introduces a fully offline Retrieval-Augmented Generation (RAG) system utilizing FAISS dense vector search and local LLMs.
- **Key Methodology:** Employs Prototyping SDLC, aligned 1000-char text chunking, and a deterministic rule-based query classifier (0.00s latency).
- **Empirical Results:** Overall RAG CPU latency reduced by 48.57% (71.69s to 36.87s) with a 0.40 similarity threshold out-of-scope grounding shield.
- **Multilingual Support:** Native support for Hausa, Arabic, French, Spanish, German, and English query handling.

SLIDE 3: BACKGROUND OF THE STUDY & PROBLEM STATEMENT

Motivation and Challenges in Academic Document Analysis

- **Manual Inefficiency:** Manual reading and keyword searching across extensive PDF/DOCX documents lead to information oversight and context loss.
- **Cloud Data Sovereignty:** Sending private research or institutional data to external cloud APIs violates data sovereignty and incurs usage fees.
- **CPU Latency Bottleneck:** Running local LLMs directly on standard CPU hardware suffers from high prefill latency during context processing.
- **Integrated Learning Support:** Existing tools lack a single platform combining document QA with Study Mode features (Summaries, Quizzes, Flashcards).

SLIDE 4: AIM AND OBJECTIVES OF THE SYSTEM

System Goals and Functional Capabilities

- **Overall Aim:** Develop a high-speed, secure, offline Intelligent Document Analysis and Learning Assistant System using local RAG.
- **Objective 1 (Multi-format Parsing):** Support seamless text ingestion from PDF, DOCX, TXT, and CSV files with page/row tracking.
- **Objective 2 (Local RAG QA):** Provide context-grounded question answering using Ollama local models (Qwen 2.5 / Gemma 2).
- **Objective 3 (Study Mode Tools):** Generate structured summaries, multiple-choice quizzes, and interactive study flashcards offline.
- **Objective 4 (Grounding Shield & Multilingual):** Enforce a 0.40 similarity threshold to block out-of-scope hallucinations and support Hausa QA.

SLIDE 5: SDLC METHODOLOGY & WHY PROTOTYPING WAS CHOSEN

Software Development Life Cycle & Architectural Rationale

- **Methodology Adopted:** Prototyping Software Development Life Cycle (SDLC) was selected over traditional Waterfall.
- **Why Prototyping? (Reason 1 - Iterative Chunking):** Allowed benchmarking different segment sizes to discover the optimal 1000-char chunking with 150-char overlap.
- **Why Prototyping? (Reason 2 - UI Refinement):** Rapidly testing and refining the collapsible sidebar, floating overlay, and Study Mode dashboards with direct user feedback.
- **Why Prototyping? (Reason 3 - Latency Optimization):** Enabled benchmark-driven iterations that introduced the 0.00s deterministic query classifier and out-of-scope grounding shield.

SLIDE 6: SYSTEM ARCHITECTURE AND DATA FLOW

End-to-End Pipeline & Component Integration

- **User Interaction Layer:** Next.js 16 responsive SPA interface communicating with backend services via REST APIs.
- **Ingestion & Extraction Layer:** Multi-format text extraction (pdf, docx, csv, txt) split into overlapping semantic segments.
- **Dense Vector Indexing:** paraphrase-multilingual-MiniLM-L12-v2 model maps text chunks into FAISS FlatIP index.
- **Intent & Search Engine:** Deterministic rule-based query classifier routes requests in 0.00s and retrieves top matching vectors.
- **Local LLM Generation:** Ollama engine executes Qwen 2.5 / Gemma 2 locally to stream grounded answers.

SLIDE 7: TECHNOLOGIES USED

Software Stack and Framework Specifications

- **Frontend Framework:** Next.js 16 (React 19) + Tailwind CSS + Lucide Icons.
- **Backend Engine:** FastAPI (Python 3.10+) + Uvicorn ASGI Server.
- **Relational Database:** SQLite (SQLAlchemy ORM for Users, Chats, Documents, Quizzes, Flashcards).
- **Vector Store:** FAISS (Facebook AI Similarity Search) with IndexIDMap and Inner Product quantizer.
- **AI Embeddings & LLM:** Sentence Transformers + Ollama Local LLM Runner (Qwen 2.5 0.5B / Gemma 2 2B).

SLIDE 8: SYSTEM DEMONSTRATION & CORE MODULES

Feature Breakdown and User Experience

- **Authentication Module:** Secure JWT token authentication with bcrypt password hashing.
- **Document Library & Upload:** Drag-and-drop file ingestion with automatic language detection.
- **Chat Interface:** Collapsible Gemini-style sidebar, floating overlay input, and streaming response viewport.
- **Study Mode Tools:** Instant automated summary extraction, multiple-choice quiz generator, and interactive flashcards.
- **Hausa Multilingual Support:** Native Hausa query detection (`ha`) with Hausa out-of-scope warning prefixes.

SLIDE 9: TESTING AND BENCHMARK RESULTS

Empirical Performance Evaluation & Latency Gains

- **RAG Execution Latency:** Overall RAG execution time on CPU reduced by 48.57% (from 71.69s down to 36.87s).
- **Classifier Speedup:** Deterministic query router reduced intent classification latency from 12.4s down to 0.00s.
- **Grounding Shield Accuracy:** 0.40 cosine similarity threshold effectively blocked 100% of out-of-scope hallucinated queries.
- **System Reliability:** All primary modules (Auth, Upload, QA, Summary, Quiz, Flashcards) passed automated verification.

SLIDE 10: CONCLUSION AND CONTRIBUTIONS

Summary of Accomplishments and Significance

- **Summary:** AuraQA successfully proves that high-performance, secure AI document intelligence is achievable locally on CPU hardware.
- **Data Sovereignty:** Eliminates all third-party cloud data transmission, ensuring 100% user data privacy.
- **Zero API Operational Cost:** Provides educational institutions with unlimited AI assistance at zero API usage fees.
- **Unified Learning Platform:** First system to integrate multi-lingual document QA, automated summarization, quizzes, and flashcards in one offline application.

SLIDE 11: KEY REFERENCES

Short Academic Literature Citations

- **Lewis et al. (2020):** Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. Advances in Neural Information Processing Systems (NeurIPS).
- **Johnson et al. (2019):** Billion-scale similarity search with GPUs. IEEE Transactions on Big Data, 7(3), 535-547 (FAISS Framework).
- **Reimers & Gurevych (2019):** Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. Proceedings of EMNLP.
- **Vaswani et al. (2017):** Attention Is All You Need. Advances in Neural Information Processing Systems (Transformer Architecture).

SLIDE 12: RECOMMENDATIONS & FUTURE WORK

Future Extensions & Defense Panel Q&A;

- **Recommendation 1 (OCR Integration):** Incorporate Optical Character Recognition (Tesseract/PaddleOCR) for scanned paper documents.
- **Recommendation 2 (Mobile App):** Develop a lightweight cross-platform mobile application powered by ONNX runtime.
- **Recommendation 3 (Voice Assistant):** Integrate local Whisper Speech-to-Text for hands-free voice interaction.
- **Acknowledgement & Q&A::** Thank you for your time and attention. I am now open to questions from the honorable panel.